MCECF: A Multiscale Complementary Enhanced Context Fusion Network for Remote Sensing Change Detection

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***Abstract*— Remote sensing change detection (RSCD) holds significant research value in remote sensing (RS) image processing. In recent years, many researchers have achieved remarkable results in RSCD tasks using methods based on convolutional neural networks (CNNs) or Transformers. Considering the limited receptive field of CNN models and the high computational cost of Transformers, many researchers have combined the two approaches, yielding promising results. However, most current RSCD-based models focus solely on change and temporal information, overlooking their complementary relationship. Additionally, some multiscale feature fusion methods emphasize enhancing individual scales while neglecting the correlations between different scales. To address the above issues, we propose a multiscale complementary enhanced context fusion (MCECF) network. The network first introduces a global–local context aggregation module (GLCAM) to capture global–local context information while extracting multilevel feature maps. Subsequently, a complementary enhancement difference module (CEDM) is employed to complementarily aggregate the captured change and temporal information of bi-temporal RS image features. To fully leverage the correlations between multiscale features, a progressive decoder comprising a supervised spatial attention (SSA) mechanism and a multiscale complementary enhanced fusion module (MCEFM) was developed. Moreover, to tackle the disparity between changed and unchanged regions, a dual-branch dynamic attention fusion module (DAFM) was designed to enhance the model's adaptability to diverse scenarios. We conducted comparative experiments on five RSCD datasets against nine state-of-the-art (SOTA) methods, and the results confirmed the effectiveness of the proposed MCECF in RSCD tasks. Our code will be made available at https://github.com/kakuqikaduo/MCECF**

***Index Terms*— Complementary enhancement, feature fusion, multiscale complementarity, remote sensing change detection (RSCD).**

I. Introduction

REMOTE sensing change detection (RSCD) aims to identify object changes between satellite images captured at different times, making it a highly challenging task. This task has significant research implications in various application areas, including object detection [1], urban planning [2], land-cover analysis [3], environmental monitoring [4], and natural disaster assessment [5]. With the advancement of satellite technology, the resolution of remote sensing (RS) images has improved, and their content has become more detailed.

The complexity of RS images, irregular geometric structures, multiscale detection objects, and environmental changes continue to pose challenges for change detection (CD) tasks. For example, Fig. 1(a)–(d) illustrates some of the challenges posed by ground realities in RSCD. These challenges include seasonal variations, content complexity, shadows, and false changes caused by lighting conditions. Given these challenges, it is essential to design an efficient CD approach capable of precisely identifying changed areas while minimizing the influence of irrelevant regions by leveraging semantic information. With advancements in traditional machine learning, numerous RSCD techniques have been introduced [6]. For example, feature differencing techniques are used to obtain binary masks through thresholding [7]. Common methods include change vector analysis (CVA) [8], principal component analysis (PCA) [9], and slow feature analysis (SFA) [10]. These pixel-based approaches mainly involve calculating, transforming, and comparing each pixel in synchronized dual-time RS images, generating change maps through appropriate thresholding or clustering algorithms. However, pixel-based methods focus only on individual pixels, neglecting the contextual information in the image. As a result, the change maps generated by these methods are often noisy and less effective. In contrast, object-based methods can focus on specific low-level semantic objects, utilizing contextual information to reduce noise in the change maps. Examples of such methods include Kullback–Leibler divergence (KL divergence) [11] and image correlation analysis [12]. However, the low-level features used in these methods are mostly artificially constructed, and a substantial gap remains when compared to the complex environments encountered in the real world. For example, issues such as pseudo-changes caused by lighting and shadows are difficult to address using the methods mentioned above.

Driven by deep learning, the performance of RSCD has undergone remarkable enhancement. At the same time, enabling end-to-end CD has significantly improved detection results. In recent years, numerous networks have been employed to address CD tasks, including convolutional neural

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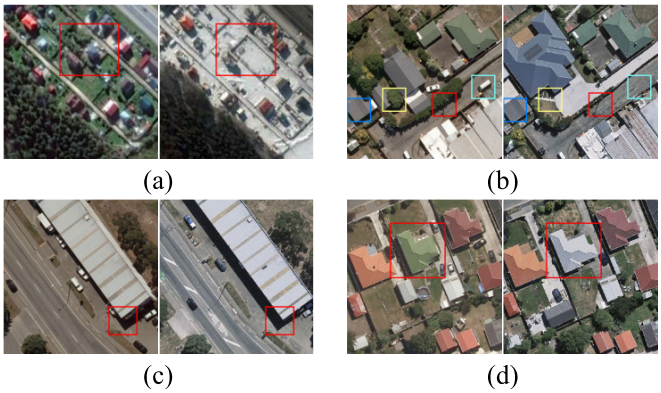


Fig. 1. Some challenges of ground realities in RSCD. (a) Red box shows the impact of seasonal variations on accurate segmentation. (b) Complexity of image content introduces certain interferences for the CD task (the blue box represents a house, the yellow box represents vegetation, the red box represents a power pole, and the cyan box represents a movable car). (c) Red box indicates the impact of false changes caused by shadows on the detection task. (d) Problem of false changes is due to varying lighting conditions.

networks (CNNs) [13], graph convolutional networks (GCNs) [14], recurrent neural networks (RNNs) [15], generative adversarial networks (GANs) [16], and Transformers [17]. Among them, CNNs have gained widespread adoption in RSCD due to their strong feature extraction abilities. Many CNN-based methods utilize the Siamese architecture to tackle the CD challenge. These approaches can be further categorized into metric-based and classification-based methods. Metric-based methods [18], [19], [20] determine changes by calculating the distance between bi-temporal image parameters in the embedding space. The "pull-in" distance is used for changed pixels, while the "push-away" distance is for unchanged pixels. Classification-based methods [21], [22], [23], [24], [25] classify each pixel in the embedding space to identify regions of change. CNN-based methods, due to their focus on local connections, struggle to capture global contextual information in RS images, potentially limiting detection performance. In contrast, Transformer-based methods [26], [27] introduce the multihead self-attention (MSA) mechanism [28], efficiently utilizing global information within images to enhance detection performance.

Although Transformer-based models have demonstrated superior performance, they also pose certain challenges. These methods typically involve high model complexity, requiring significant memory and high floating-point operations per second (FLOPs). Due to the complexity of RSCD tasks, these methods struggle to effectively integrate local contextual information and accurately segment fine-grained irregular edge changes. On the other hand, most current RSCD-based models focus only on change and temporal information, overlooking their complementarity. Additionally, some multiscale feature fusion methods rely on enhancing individual scales while neglecting the relationships between scales.

To overcome the limitations mentioned above, we introduce an efficient Siamese-based RSCD framework. Specifically, a multiscale complementary enhanced context fusion (MCECF) network. Our design focuses on three key components. First, in the encoder, a global–local context aggregation module (GLCAM) is employed to capture global–local contextual information while extracting multiscale features. Second, considering the complementarity between temporal and change features, we design a complementary enhancement difference module (CEDM). Third, a supervised spatial attention (SSA) mechanism and multiscale complementary enhanced fusion module (MCEFM) are used for progressive decoding. Considering the imbalance between change and nonchange regions, we design a dynamic attention fusion module (DAFM) to enhance sensitivity to change regions, resulting in higher-quality bi-temporal change maps. The key contributions of our work are outlined as follows.

1) We propose a Siamese-based RSCD network (MCECF) capable of capturing both local and global contextual information. Furthermore, it explores the relationships between temporal and change features to refine and enhance multiscale features, producing high-quality CD maps. Experiments on five datasets demonstrate that MCECF achieves SOTA performance with strong robustness.
2) The encoder part consists of an encoder block formed by GLCAM, which captures global–local context information. The feature maps output at each stage are processed by CEDM, which learns time and change information through a dual-branch structure. To leverage the relationships between multiscale features, we designed a progressive decoder comprising an SSA mechanism and an MCEFM.
3) Considering the imbalance between change and nonchange regions, we designed a DAFM with a dual-branch structure. This module is applied to the MCEFM in the decoder and the final change map generation stage to enhance the model's adaptability across different scenarios.

The structure of this article is organized as follows. An overview of related work on RSCD is provided in Section II. The details of our MCECF implementation are introduced in Section III. Section IV presents the experiments and corresponding discussion. Finally, Section V provides a concise summary to conclude this article.

II. Related Work

A. CNN-Based CD Models

CNNs can automatically learn powerful feature representations with the help of large-scale manually annotated data samples. Existing RSCD models are mostly based on the Siamese architecture [18], [19], [29], [30]. Compared to single-branch models [31], dual-branch models process bi-temporal RS images simultaneously for feature extraction, resulting in more efficient change map generation. Liu et al. [32] proposed an effective framework, SRCDNet, which uses multiscale convolutional attention features and learns bi-temporal feature differences through adversarial learning. Chen et al. [33] developed a dual encoder–decoder network (FCCDN) that employs dense connections and self-supervised learning techniques to achieve favorable results in RSCD tasks. In addition to constructing Siamese architectures to

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improve RSCD efficiency, some researchers also adopt multiscale feature fusion strategies to combine the advantages of high-level and low-level features from different layers. TFI-GR [34] designed a temporal feature interaction and guidance refinement network that aggregates temporal change information from multilevel features. A2Net, proposed by Li et al. [23], improves the representation capability of temporal features extracted from low, mid, and high levels by incorporating a neighbor aggregation module. Additionally, considering the spatial context information of bi-temporal RS images, Huang et al. [13] developed SEIFNet, a method designed to extract contextual information from bi-temporal RS images through spatiotemporal difference enhancement and adaptive context fusion. The approach incorporates a difference module that accounts for both local and global context, while combining features from both high and low levels to enhance decoding accuracy, ultimately resulting in precise and efficient change map generation. To focus on the complementarity between change and temporal information in RSCD, Lu et al. [35] proposed CTSFFNet, a cross-temporal symmetric feature fusion network that integrates temporal and change features.

B. Transformer-Based CD Models

In contrast, transformers use self-attention mechanisms to model long-range dependencies. By interacting with the intrinsic information of features, transformers capture the inherent correlations between them, enabling Transformer-based CD models [36], [37], [38], [39] to focus on global information and generate higher-quality change maps. ChangeFormer [17] integrates a hierarchical Transformer encoder with a multilayer perceptron (MLP) decoder, effectively capturing the multiscale long-term details essential for precise CD. Sun et al. [40] developed an RSCD method that leverages a spatial focus dual-temporal interaction network (SFBI-Net). This method performs global interaction of bi-temporal features within the same network layer and captures the spatiotemporal relationships of these features by supplementing local information, enabling distinct representation of bi-temporal features. To address class imbalance and pseudo-change issues caused by lighting in RSCD tasks, Li et al. [41] proposed a spatiotemporal attention network (STADE-CDNet) based on difference enhancement, which extracts change regions from the difference map through a CD difference enhancement module, enhancing the network's learning capacity and mitigating the impact of class imbalance. Noman et al. [42] proposed an effective local and global context aggregation network (ELGC-Net) that significantly enhances the performance of CD tasks in RS. This method integrates a specialized local–global context aggregation module into the encoder, which plays a pivotal role in capturing both local and global contextual information.

C. Attention Mechanism in CD

Existing CD models frequently employ attention mechanisms to direct the model's focus toward regions of interest that display significant changes, thereby improving detection accuracy and efficiency [19], [43], [44]. Examples include spatial attention [45], self-attention [36], and cross-attention [37]. DSAMNet [19] introduces a convolutional attention module (CBAM) within the metric module, combining spatial and channel attention to enhance feature discriminability. Chen and Shi [18] presented a pyramid spatiotemporal attention module (PAM) incorporating a self-attention mechanism, designed to capture multiscale features and intricate details from dynamic regions, thereby enhancing the model's ability to detect changes more effectively. SEIFNet [13] incorporates coordinated attention (CA) [46], capturing long-range dependencies across channels, time, and space dimensions. To mitigate the side effects of cross-channel relationship modeling based on channel reduction, Ouyang et al. [47] introduced a novel and efficient multiscale attention mechanism that maintains channel information while reducing computational overhead. By partitioning the channel dimension into multiple subfeatures, the method ensures an even distribution of spatial semantic features within each feature group, leading to improved computational efficiency and enhanced model performance.

Due to the complexity of the RSCD task, many existing methods struggle to effectively integrate global–local contextual information, which limits the precise segmentation of certain details. Furthermore, most current RSCD models fail to consider the complementarity between temporal and change information. Additionally, some methods that rely on multiscale feature fusion tend to focus on enhancing individual scales without considering the interrelationships between them. To address these issues, we developed an effective CD model (MCECF) to generate higher-quality change maps.

III. Proposed Method

A. Overall Framework

The overall architecture of the MCECF network is shown in Fig. 2. The proposed network architecture consists of three key components: the encoder, the difference module, and the decoder, each playing a crucial role in the overall CD process. The encoder uses a Siamese architecture to take the dual-temporal remote sensing images as input x_{pre} and x_{post} outputs multiscale feature maps at four stages. Feature maps at each stage are downsampled via patch embedding, and then, using the encoder block, features at different scales are extracted across the four stages. GLCAM captures global–local context information. At each stage, the CEDM learns the correlation between temporal and change information to enhance detailed features while highlighting the differences between the dual-temporal RS images. Finally, we construct a progressive decoder using MCEFM formed by DAFM and SSA, which refines and enhances fine details by combining the advantages of high-level and low-level features, achieving complementary information fusion between multiscale features to generate high-quality change maps. We also added DAFM after the encoder to enhance the model's ability to focus differently on change and nonchange regions, thereby improving the detection of small changes.

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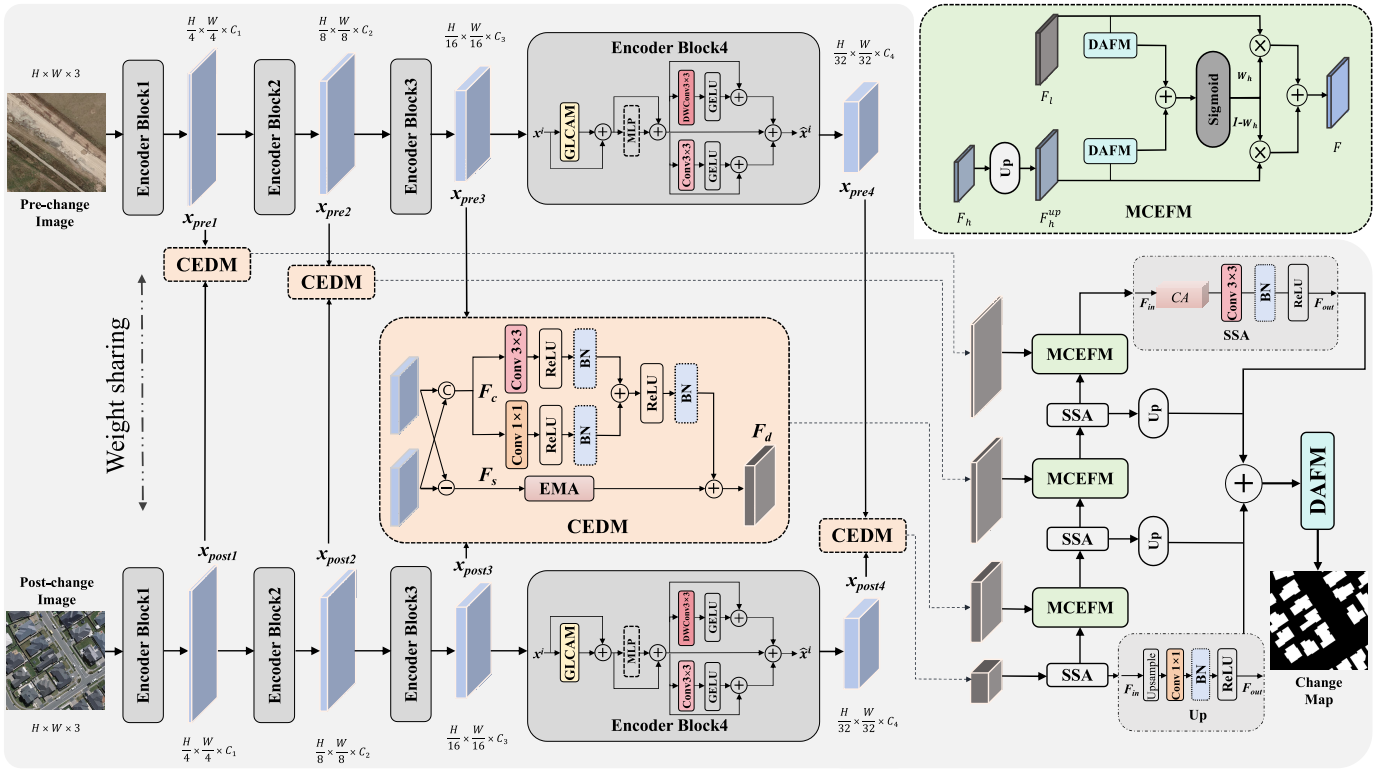


Fig. 2. Overall framework of MCECF. In the feature extraction stage, we use patch embedding, with feature maps generated at four stages having dimensions of 64 × 64, 32 × 32, 16 × 16, and 8 × 8, respectively. The framework of MCEFM is shown in the top-right corner, with detailed structural diagrams of MCEFM, GLCAM, and DAFM presented in Section III.

B. Global–Local Context Aggregation Module

The proposed encoder includes GLCAM, MLP, and a dual-branch convolutional layer, as shown in Encoder Block 4 of Fig. 2. First, residual connections are introduced by GLCAM and MLP, where GLCAM captures global–local context information after channel segmentation, and MLP integrates cross-channel features to enhance nonlinear representations. Next, 3 × 3 depthwise convolutions and standard convolutions enhance the richness of feature representations. The proposed encoder captures both global and local context information, generating multiscale feature maps at four stages. Inspired by the ELGC-Net proposed by Noman et al. [42], we divide the input features into different groups based on channels and then apply different operations on the channels in each group, as shown in Fig 3. Specifically, the input feature maps are first segmented by channels and fed into the convolutional branch and the attention branch. The convolutional branch enhances local information, while the attention branch captures global context information. Let the input feature at the ith stage be $x^i \in \mathbb{R}^{H^i \times W^i \times C^i}$ (where H^i, W^i, and C^i represent height, width, and channels, respectively). The input is split by channels to obtain x_1^i, $x_2^i \in \mathbb{R}^{H^i \times W^i \times (C^i/2)}$, that is,

$$x_1^i, x_2^i = \text{Split}(x^i). \tag{1}$$

1) In the convolutional branch, parallel branches are used to capture local spatial information while integrating global context information. The input x_1^i undergoes 3 × 3 depthwise convolution to capture local context information and adaptive average pooling to extract global context as a supplement. A 1 × 1 pointwise convolution is applied after average pooling to integrate cross-channel information. The features from the two branches are then multiplied element-wise, and the result is passed through a GELU activation layer. The fusion of the two branches $\bar{x}_1^i$ allows the convolutional branch to enhance local features while preserving global information. The implementation of the convolutional branch is given by the following equation:

$$\bar{x}_1^i = \text{DWConv3}(x_1^i) \times \text{Conv1}(\text{AvgPool}(x_1^i)). \tag{2}$$

2) In the attention branch, we first split x_2^i into features Q^i, K^i, V^i, and L^i and apply 1 × 1 convolution to each. Where, Q^i, K^i, and V^i are used as query, key, and value, respectively, while L^i retains local context information for the final fusion. Adaptive average pooling is applied to Q^i and K^i to enhance attention mechanism expression while reducing parameters and computation in subsequent attention calculations, resulting in global features $\bar{Q}^i$ and $\bar{K}^i$. Transposed attention (G) is then applied between $\bar{Q}^i$ and $\bar{K}^i$, where the feature channel count of $\bar{Q}^i$ and $\bar{K}^i$ is $C/4$. Compared to standard self-attention, our attention branch achieves higher model efficiency while capturing global context information with greater computational efficiency. The attention feature representation can be given by (3), where $\bar{Q}^i$, $\bar{K}^i \in \mathbb{R}^{(H^i/2)\times(W^i/2)\times(C^i/4)}$, $\bar{V}^i \in \mathbb{R}^{H^i \times W^i \times (C^i/4)}$ and

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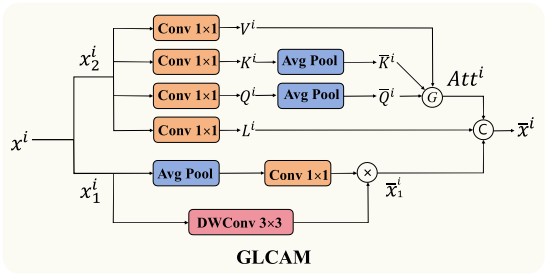


Fig. 3. Illustration of GLCAM. A 1×1 convolution is used for linear projection to facilitate the subsequent attention score calculation. Transposed attention (G) is applied between $\bar{Q}^i$ and $\bar{K}^i$, where the feature channel count of $\bar{Q}^i$ and $\bar{K}^i$ is $C/4$. Compared to standard self-attention, the proposed Att^i is more efficient.

$\left(\bar{K}^i\right)^T$ denote the transposed forms of $\bar{K}^i$, respectively,

$$\mathrm{Att}^i = \left\{\mathrm{Softmax}\left[\bar{Q}^i \times \left(\bar{K}^i\right)^T\right]\right\} \times V^i. \tag{3}$$

Finally, the local context information enhanced by the convolutional branch and the global context information captured by the attention branch are fused using a cascaded operation, as given by the following equation:

$$\bar{x}^i = \mathrm{Cat}\left(\bar{x}_1^i, L^i, \mathrm{Att}^i\right). \tag{4}$$

C. CEDM

Currently, many RSCD methods focus solely on capturing temporal and change information independently, neglecting their complementary relationship. Inspired by CTSFFNet [35], as shown in Fig. 2, we proposed the CEDM to capture temporal information and change-related features in bi-temporal RS images while considering their complementarity. The subtraction branch computes the difference between the inputs and, followed by the EMA attention mechanism [47] to capture complementary information across channels. The connection branch combines image features from the two temporal points to ensure the model focuses on temporal information when processing features. The combined features are then processed with convolutions of varying kernel sizes to improve the model's understanding of feature changes at different scales. Finally, the outputs of the subtraction and connection branches are summed to obtain the difference-enhanced features. CEDM effectively enhances the sensitivity to change regions in RS images by leveraging multiscale difference enhancement and the complementarity of temporal and change information, enabling more accurate capture and processing of change features. The overall implementation of CEDM can be represented by the following equation:

$$F_c = \mathrm{Cat}\left(x_{\mathrm{pre}}, x_{\mathrm{post}}\right) \tag{5}$$

$$F_c' = \mathrm{BN}(\mathrm{ReLU}(\mathrm{BN}(\mathrm{ReLU}(\mathrm{Conv3}(F_c))) + \mathrm{BN}(\mathrm{ReLU}(\mathrm{Conv1}(F_c))))) \tag{6}$$

$$F_s' = \mathrm{EMA}\left(\left|x_{\mathrm{pre}} - x_{\mathrm{post}}\right|\right) \tag{7}$$

$$F_d = F_c' + F_s'. \tag{8}$$

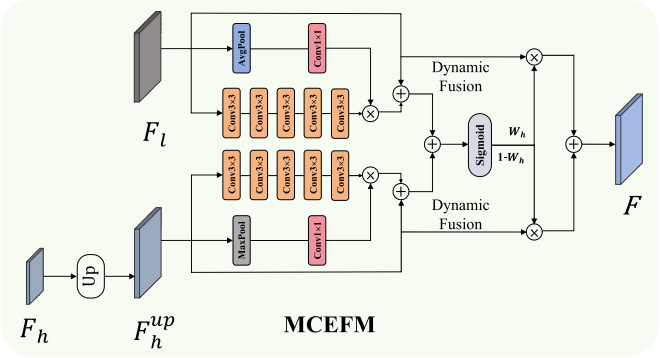


Fig. 4. Illustration of MCEFM. The features F_l and F_h^{up} are passed through DAFM and then summed to obtain the fusion weight W_h. The weights W_h and $1 - W_h$ are multiplied with the low-level and high-level features, respectively, to complement the advantages of both.

D. Progressive Decoder (MCEFM and SSA)

In RSCD tasks, multiscale feature fusion is a key method for improving model performance. Many traditional methods perform feature fusion using simple concatenation or averaging operations, which may result in information loss. Some methods, in pursuit of generating more accurate change maps, excessively focus on high-frequency information while neglecting the correlation between high-frequency and low-frequency information. To address these challenges, we propose the MCEFM, which effectively integrates the global context information from low-level features with the intricate details captured by high-level features, as illustrated in Fig. 4. First, bilinear interpolation is used for upsampling the high-level features, followed by batch normalization (BN) [48] and ReLU to reduce the channel dimensions of the upsampled features, ensuring the same resolution as the low-level features for seamless fusion. Next, the proposed adaptive DAFM is used to calculate the weights for high-frequency information. These weights and their complements (1 minus the weights) are multiplied with low-level and high-level features, respectively, to combine their advantages. The process is represented by the following equations:

$$W_h = \mathrm{sigmoid}\left[\mathrm{DAFM}(F_l) + \mathrm{DAFM}\left(F_h^{\mathrm{up}}\right)\right] \tag{9}$$

$$F = W_h \times F_l + (1 - W_h) \times F_h^{\mathrm{up}}. \tag{10}$$

We also propose an SSA mechanism to help the model better leverage label information, improving its spatial feature representation and focusing more effectively on change regions, as shown in Fig 2. SSA consists of coordinate attention (CA) [46] and a serial combination of the 3×3 convolutional layer, subsequently followed by BN and a ReLU activation function. The SSA can be given by the following equation:

$$F_{\mathrm{out}} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv3}(\mathrm{CA}(F_{\mathrm{in}})))). \tag{11}$$

E. Dynamic Attention Fusion Module

RS images typically contain abundant geographical information and diverse scenes. In RSCD tasks, change regions

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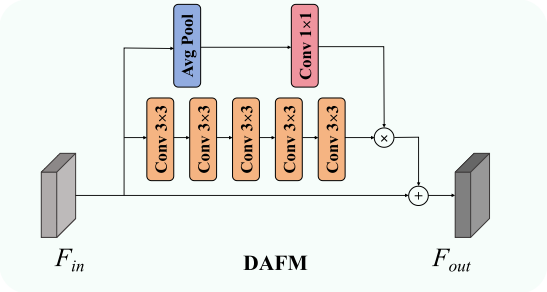


Fig. 5. Illustration of DAFM. The proposed DAFM consists of an attention branch and a convolutional branch. The attention branch enhances the feature representation of important regions, while the convolutional branch captures multiscale and multilevel information, ensuring the model can learn complex feature relationships. In the convolutional branch, each 3×3 convolution layer is followed by a ReLU activation function to enhance feature representation.

generally represent only a small portion of the entire image. Considering the imbalance between change and nonchange regions, we propose the DAFM, as shown in Fig. 5. Leveraging an adaptive attention mechanism, DAFM dynamically adjusts the focus on features, effectively suppressing irrelevant information and enhancing model robustness. We integrate DAFM into MCEFM to further enhance multiscale feature fusion. DAFM consists of an attention branch and a convolutional branch. The attention branch applies adaptive average pooling and 1×1 convolution to weight feature maps, enhancing the representation of important regions. This mechanism allows the model to dynamically adjust focus in response to the input. The convolutional branch stacks five layers of 3×3 convolutions to capture multiscale and hierarchical information, enabling the model to learn complex feature relationships.

IV. Experiments

A. Datasets and Evaluation Scheme

In our experiments, we validated the effectiveness of the proposed method using five datasets.

1) *LEVIR-CD [18]:* This dataset consists of 637 images from Google Earth for CD, primarily focusing on building changes. Each image has dimensions of 1024×1024 pixels and a resolution of 0.5 m. The original images were cropped to 256×256 pixels for the experiments. Of these, 7120 images are used for training, 1024 for validation, and 2048 for testing.
2) *WHU-CD [49]:* The WHU dataset focuses on buildings. The original RS images have a spatial resolution of 0.2 m and dimensions of 32507×15354 pixels. Similar to [39], the images were cropped into 256×256 patches and randomly partitioned into 5947 patches for training, 743 for validation, and 744 for testing.
3) *CDD-CD [50]:* This dataset includes objects such as seasonal changes, cars, roads, and buildings. Following the original study, we used 10 000/3000/3000 images for training, validation, and testing, respectively.
4) *BANDON-CD [51]:* This dataset comprises images with a spatial resolution of 0.6 m and dimensions of 2048×2048 pixels. The images are sourced from multiple platforms, including Google Earth, Microsoft Virtual Earth, and ArcGIS, and primarily focus on buildings. To validate our method, the images were cropped into 1024×1024 patches, and resized to 256×256 pixels, resulting in 6756 images for training, 808 for validation, and 828 for testing.
5) *CLCD-CD [52]:* Unlike the previous four datasets, CLCD focuses primarily on cropland changes. It contains 600 pairs of 512×512-pixel images, with a resolution ranging from 0.5 to 2 m/pixel. The original dataset was cropped into 256×256 patches, yielding 1440 images for training, 480 for validation, and 480 for testing.

Evaluation Scheme: To quantitatively assess the effectiveness of the proposed method, we use five widely used metrics to evaluate the CD results: precision (Pre), recall (Rec), F1-score (F1), intersection over union (IoU), and overall accuracy (OA). The definitions are as follows:

$$\text{Pre} = \text{TP}/(\text{TP} + \text{FP}) \tag{12}$$

$$\text{Rec} = \text{TP}/(\text{TP} + \text{FN}) \tag{13}$$

$$\text{F1} = 2 \times \text{Pre} \times \text{Rec}/(\text{Pre} + \text{Rec}) \tag{14}$$

$$\text{IoU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}) \tag{15}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{16}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively.

B. Experimental Settings

We trained the model using the PyTorch framework on an NVIDIA GeForce RTX 3090 GPU with 24GB of memory. During the training process, various data augmentation techniques, including flipping, cropping, and blurring, were employed. We used the AdamW optimizer with β values of (0.9, 0.999) and weight decay of 0.01. The initial learning rate was set to 0.0001, and the batch size was 8. The model was trained for 150 epochs, and the input image size for all datasets was 256×256.

Loss Function: During training, we used a combination of Dice loss and binary cross-entropy (BCE) loss to evaluate the network's performance

$$L_{\text{Dice}}(y, \hat{y}) = 1 - \frac{2y\hat{y} + 1}{y + \hat{y} + 1} \tag{17}$$

$$L_{\text{BCE}}(y, \hat{y}) = -\left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})\right] \tag{18}$$

$$\text{Loss} = \lambda L_{\text{Dice}}(y, \hat{y}) + (1 - \lambda) L_{\text{BCE}}(y, \hat{y}) \tag{19}$$

where y represents the ground-truth label, and $\hat{y}$ denotes the predicted change result, λ with a value of 0.25.

C. Comparison With SOTA Methods

To assess the effectiveness of the proposed method, we perform a comparative analysis against nine SOTA approaches recently introduced in the field. TFI-GR [34] utilizes feature interaction to capture temporal difference information at different levels. Both A2Net [23] and ELW_CDNet [53] are lightweight networks designed to reduce model

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TABLE I

QUANTITATIVE COMPARISON OF CD RESULTS ON THE LEVIR-CD, WHU-CD, AND CDD-CD DATASETS IS PRESENTED, WITH THE BEST RESULTS HIGHLIGHTED IN RED AND THE SECOND-BEST IN BLUE

Methods	**LEVIR-CD(%)**					**WHU-CD(%)**					**CDD-CD(%)**				
	IoU	F1	Rec	Pre	OA	IoU	F1	Rec	Pre	OA	IoU	F1	Rec	Pre	OA
TFI-GR [34]/2022	86.02	92.00	91.10	92.96	98.49	78.90	86.91	85.57	88.36	98.15	94.08	96.89	96.29	97.52	98.72
ChangeFormer [17]/2022	86.23	92.14	90.42	94.03	98.54	89.33	94.08	92.94	95.30	99.16	83.92	90.79	89.05	92.76	96.33
A2Net [23]/2023	88.18	93.38	90.96	96.15	98.79	92.81	96.15	97.05	95.28	99.43	92.85	96.21	95.48	96.98	98.45
ELW_CDNet [53]/2023	81.30	89.68	88.64	90.75	98.96	87.56	93.37	92.20	94.56	99.50	90.44	94.98	94.29	95.67	98.71
STADE-CDNet [41]/2024	87.61	93.02	90.53	95.89	98.73	92.11	95.74	94.33	97.26	99.40	82.70	89.95	85.06	97.17	96.34
SFBI-Net [40]/2024	89.28	94.06	93.23	94.94	98.87	92.32	95.87	94.76	97.02	99.41	95.48	97.66	97.56	97.76	99.03
ELGC-Net [42]/2024	89.91	94.45	94.25	94.66	98.93	91.18	95.20	95.10	95.30	99.30	93.83	96.76	96.47	97.05	98.66
SEIFNet [13]/2024	89.41	94.14	93.46	94.85	98.89	92.30	95.86	95.86	95.86	99.39	95.12	97.46	97.14	97.80	98.95
ScratchFormer [54]/2024	89.98	94.52	93.88	95.17	98.98	89.72	94.32	93.97	94.69	99.18	95.81	97.84	97.86	97.81	99.10
MCECF(Ours)	90.51	94.81	94.17	95.47	99.01	93.55	96.58	96.14	97.03	99.51	96.91	98.42	98.55	98.29	99.34

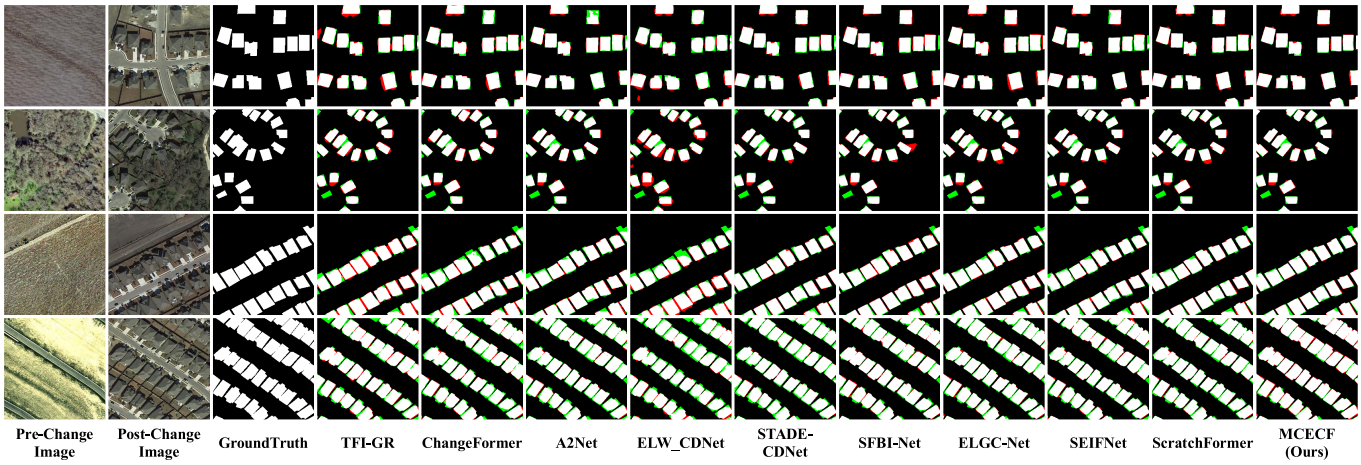


Fig. 6. Qualitative comparison of our method with the most recent SOTA techniques is performed on the LEVIR-CD dataset. To enhance clarity, we use white, black, red, and green to denote TPs, TNs, FPs, and FNs, respectively.

complexity and improve inference speed. ChangeFormer [17] is a Transformer-based framework designed for CD. STADE-CDNet [41], SFBI-Net [40], ELGC-Net [42], and ScratchFormer [54] are all transformer-based models. Among them, STADE-CDNet and SFBI-Net enhance detection performance by amplifying spatiotemporal differences, ELGC-Net estimates change regions by utilizing global–local contextual information, and ScratchFormer captures inherent features of CD data through hybrid sparse attention. SEIFNet [13] achieves adaptive context fusion through spatiotemporal enhancement and interlayer fusion. To ensure fairness in the experiments, we retrained and tested these methods using the official code provided in the original papers. The hyperparameters were set as per the original implementations, and the experimental settings were consistent with those of our proposed method.

1) Quantitative Evaluation: Table I provides a quantitative comparison of the proposed method with SOTA approaches on the LEVIR-CD, WHU-CD, and CDD-CD datasets. Furthermore, to further validate the effectiveness of our method, the CLCD-CD and BANDON-CD datasets were also included for evaluation, as presented in Table II.

a) Table I: On the LEVIR-CD dataset, our method outperforms the compared SOTA methods in IoU, F1, and OA, achieving 90.51%, 94.81%, and 99.01%, respectively. Although A2Net achieves the highest precision score of 96.15% among all compared methods, its recall is notably lower than most, at only 90.96%, resulting in a lower overall F1 score. ELGC-Net achieves the highest recall score of 94.25%, while our MCECF ranks second in the recall. However, our method achieves the highest overall F1 score, indicating that it maintains high precision while minimizing the FN rate. On the WHU-CD dataset, our proposed method outperforms existing methods, achieving 93.55%, 96.58%, and 99.51% for IoU, F1, and OA, respectively. Notably, our method ranks second in both precision and recall, while our precision of 97.03% is slightly lower than STADE-CDNet's 97.26%. Our recall is 0.91% lower than A2Net, and although our overall F1 score is the highest, this also indicates that our method still requires improvement on the WHU-CD dataset. On the CDD-CD dataset, our method outperforms the SOTA methods across all five metrics—IoU, F1, recall, precision, and OA—achieving 96.91%, 98.42%, 98.55%, 98.29%, and 99.34%, respectively. These results illustrate that our method delivers superior detection performance compared to the existing SOTA approaches.

b) Table II: On the BANDON-CD dataset, our method achieved the highest scores of 64.78%, 73.50%, and 69.42%

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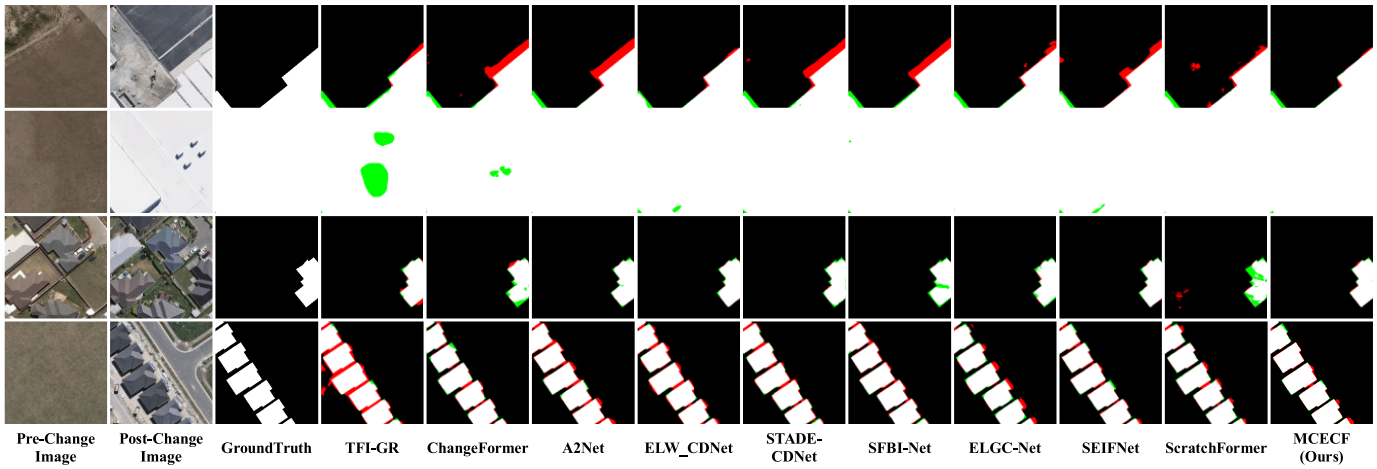


Fig. 7. Qualitative comparison of our method with the most recent SOTA techniques is performed on the WHU-CD dataset. To enhance clarity, we use white, black, red, and green to denote TPs, TNs, FPs, and FNs, respectively.

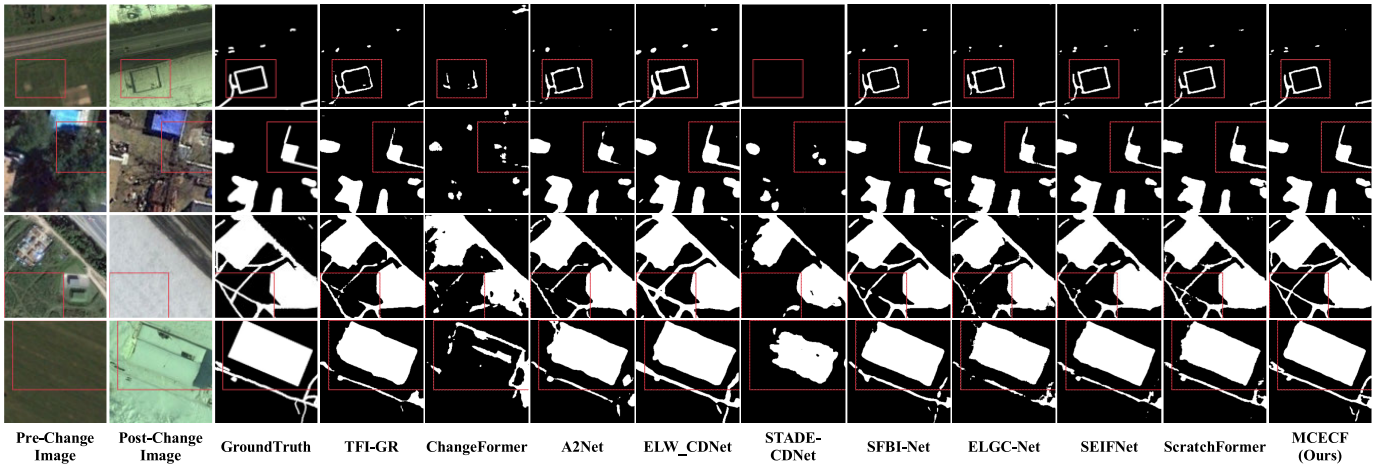


Fig. 8. Qualitative comparison between our method, and the latest SOTA techniques is carried out on the CDD-CD dataset. Areas highlighted in red denote regions where our method achieves better detection performance compared to other SOTA methods.

TABLE II

QUANTITATIVE COMPARISON OF CD RESULTS ON THE BANDON-CD AND CLCD-CD DATASETS IS PRESENTED, WITH THE BEST RESULTS HIGHLIGHTED IN RED AND THE SECOND-BEST IN BLUE

Methods	**BANDON-CD(%)**					**CLCD-CD(%)**				
	IoU	F1	Rec	Pre	OA	IoU	F1	Rec	Pre	OA
TFI-GR [34]/2022	48.81	49.40	50.00	48.81	97.61	68.45	78.48	77.68	79.34	94.24
ChangeFormer [17]/2022	56.62	62.87	58.71	76.79	97.70	65.99	76.17	75.89	76.46	93.51
A2Net [23]/2023	64.66	73.35	68.84	81.00	97.88	65.95	75.84	71.37	83.57	94.53
ELW_CDNet [53]/2023	45.06	62.13	61.70	62.56	97.20	57.14	72.72	71.45	74.04	96.01
STADE-CDNet [41]/2024	53.55	57.98	50.31	84.76	97.75	67.49	77.61	77.45	77.79	93.87
SFBI-Net [40]/2024	61.71	69.81	64.97	79.95	97.89	68.19	78.16	75.87	81.02	94.48
ELGC-Net [42]/2024	64.46	73.11	68.51	81.11	98.00	71.10	80.85	79.83	81.96	94.90
SEIFNet [13]/2024	64.46	73.13	68.98	79.86	97.95	70.94	80.72	80.06	81.42	94.80
ScratchFormer [54]/2024	62.30	70.53	65.41	81.49	97.95	69.34	79.29	78.31	80.35	94.48
MCECF(Ours)	64.78	73.50	69.42	80.00	97.96	73.44	82.86	82.22	83.53	95.37

for IoU, F1, and recall, respectively. In terms of OA, our method is 0.04% lower than ELGC-Net, but it is 4.76% lower than the highest precision score from STADE-CDNet. This indicates that the detection accuracy of our model on the BANDON-CD dataset still needs improvement. On the CLCD-CD dataset, our method achieved the highest scores in IoU, F1, and recall, with values of 73.44%, 82.86%, and 82.22%, respectively. Our method is 0.04% lower than A2Net in precision and 0.64% lower than ELW_CDNet in OA. This demonstrates that our MCECF method also achieves good detection performance when applied to land-use CD, showing strong generalization ability.

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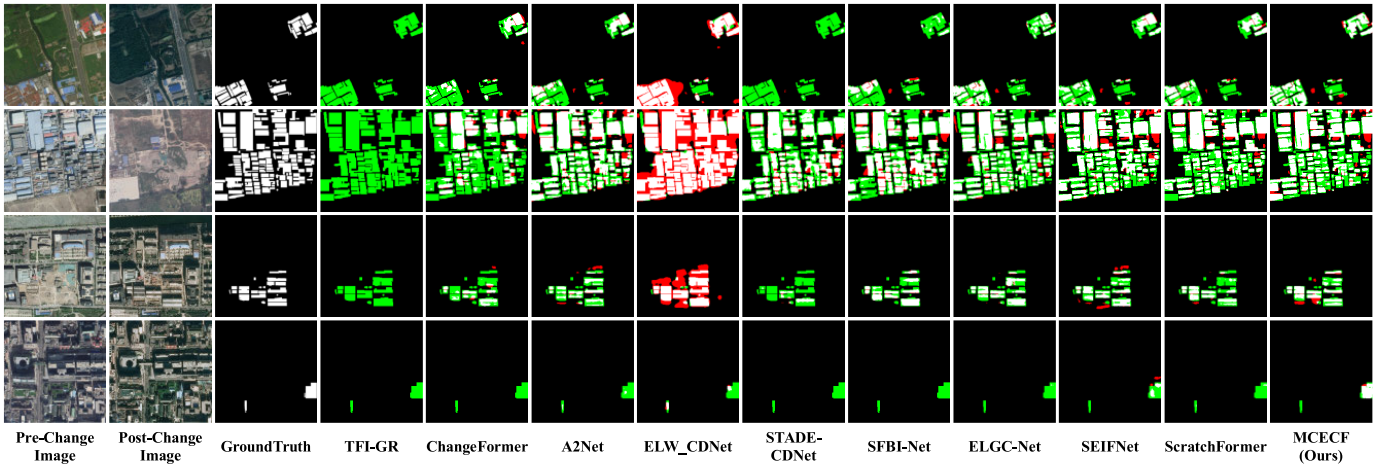


Fig. 9. Qualitative comparison of our method with the most recent SOTA techniques is performed on the BANDON-CD dataset. To enhance clarity, we use white, black, red, and green to denote TPs, TNs, FPs, and FNs, respectively.

2) Qualitative Evaluation: Figs. 6–10 present the CD results of the proposed method and other comparative methods, visualized across the LEVIR-CD, WHU-CD, CDD-CD, BANDON-CD, and CLCD-CD datasets. For each dataset, we selected four pairs of images with different temporal changes for demonstration.

The CD targets in Figs. 6, 7, and 9 are buildings. In Fig. 6, all methods achieved good detection performance; however, it is noteworthy that when buildings transition from sparse to more compact arrangements, our method outperforms the others. As shown in the fourth row of Fig. 6, our method exhibits a lower false-negative rate compared to the other methods, leading to a more regular edge in the generated change map. This demonstrates the effectiveness of our proposed MCEFM module's strategy of complementary aggregation of high-level and low-level features. However, in the second row, our method, like the other methods, fails to detect the change in the building at the lower left corner. This might be due to the high similarity between the building's surface and its surrounding environment, causing the model to miss the change. In contrast to Fig. 6, the detection targets in Fig. 7 are also buildings, but the challenges they present are different. In the second and third rows of Fig. 7, the reflective surfaces of the buildings, due to lighting conditions, affect the detection performance of the model. In the second-row results, TFI-GR, ChangeFormer, ELW_CDNet, and SEIFNet all suffer from FNs due to the reflection, whereas our method remains unaffected. In the third row, except for ScratchFormer, which experiences some interference leading to a higher false-negative rate, the other methods perform similarly. Overall, compared to the other methods, our method generates higher-quality change maps. When compared to the first two datasets, the BANDON dataset presents greater complexity and a higher final resolution, resulting in poorer detection performance across all methods, including ours. However, our method still achieves higher overall detection accuracy. For instance, in the fourth row of Fig. 9 (bottom-right corner), only our model effectively detects changes in the building, while all other methods, except for SEIFNet, fail to detect the changes. This demonstrates that although our model performs less effectively on this dataset, it still outperforms existing state-of-the-art (SOTA) methods.

The detection targets in the CDD-CD dataset of Fig. 8 are more diverse. By comparing the detection results highlighted in red boxes, our method outperforms other SOTA methods in accurately detecting changes in the target regions. Furthermore, it maintains a strong connection between the region boundaries and preserves the internal details and edge accuracy, resulting in better overall detection performance compared to the other methods.

Fig. 10 focuses on cultivated land changes. In the first two rows of results, our method achieves significantly lower false-negative rates, while all other methods exhibit substantial FNs. In the third row, although our method produces the best overall results, it fails to detect the cultivated land changes at the top, which evolved from a building. This indicates that while our method shows a certain degree of generalization ability, there is still room for improvement.

D. Ablation Study

To assess the effectiveness of the proposed modules, we carried out ablation studies on three datasets. For clarity, we use only IoU, F1, and OA as evaluation metrics. We performed the ablation by removing each module and comparing it with MCECF. The ablation results for each module are shown in Table III.

1) Ablation of GLCAM: We capture global–local context information through the GLCAM encoder, achieving adaptive integration of context information, and output four features at different scales. To evaluate the effectiveness of GLCAM, we replace our encoder block with a Transformer block that includes self-attention to output feature maps at the same scale. As shown in Table III, when the standard self-attention Transformer replaces our encoder, the IoU, F1, and OA detection accuracy of MCECF decreases across the LEVIR-CD, WHU-CD, and CDD-CD datasets, with a decrease of 2.78%, 1.50%, and 0.62%, respectively, on the CDD-CD dataset. This demonstrates that our proposed GLCAM effectively integrates

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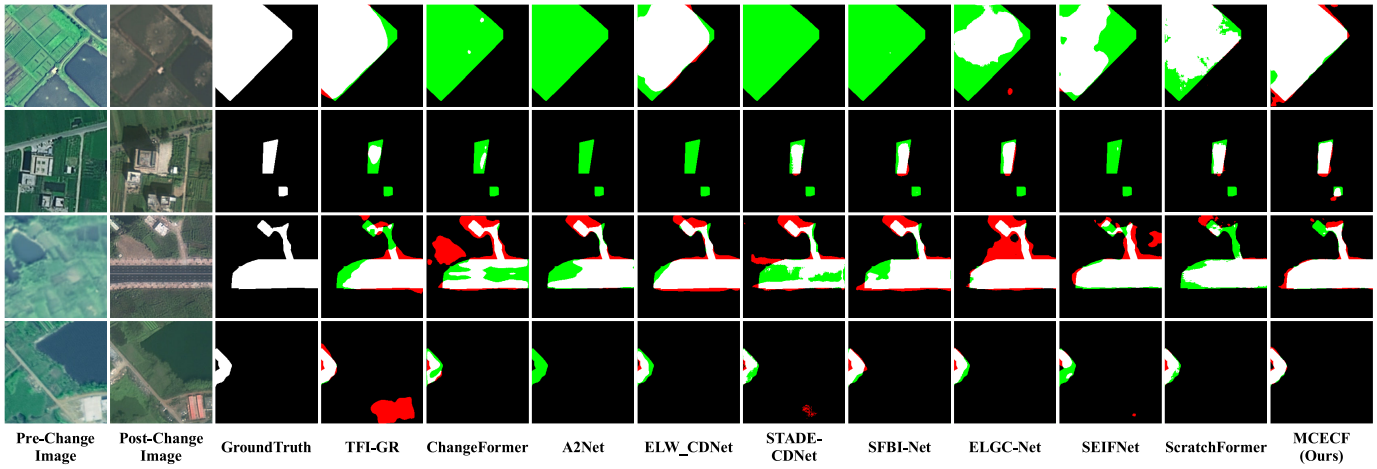


Fig. 10. Qualitative comparison of our method with the most recent SOTA techniques is performed on the CLCD-CD dataset. To enhance clarity, we use white, black, red, and green to denote TPs, TNs, FPs, and FNs, respectively.

TABLE III

QUANTITATIVE COMPARISON OF THE CD RESULTS FROM THE ABLATION EXPERIMENTS OF EACH MODULE ON THE LEVIR-CD, WHU-CD, AND CDD-CD DATASETS

Methods	LEVIR-CD(%)			WHU-CD(%)			CDD-CD(%)		
	IoU	F1	OA	IoU	F1	OA	IoU	F1	OA
MCECF	**90.51**	**94.81**	**99.01**	**93.55**	**96.58**	**99.51**	**96.91**	**98.42**	**99.34**
w/o GLCAM	89.12	93.97	98.86	91.20	95.21	99.29	94.13	96.92	98.72
w/o CEDM	90.08	94.55	98.87	93.08	96.30	99.47	96.51	98.21	99.25
w/o MCEFM	90.28	94.65	98.98	92.64	96.05	99.43	96.79	98.35	99.31
w/o SSA	90.14	94.59	98.96	90.77	94.96	99.25	96.80	98.36	99.32
w/o DAFM	90.05	94.54	98.96	91.97	95.66	99.36	96.34	98.11	99.21

context information, thereby improving the model's detection performance.

2) Ablation of CEDM: CEDM captures change areas and suppresses certain interferences by complementing time and change information. We replace our CEDM with absolute subtraction of the dual-time inputs after feature extraction to further validate its effectiveness. After replacing CEDM with the simple absolute subtraction operation, the model's detection accuracy declined. Specifically, on the LEVIR-CD dataset, IoU, F1, and OA decreased by 0.43%, 0.26%, and 0.14%, respectively. This indicates that our proposed CEDM complements time and change information, thereby improving the model's sensitivity to change areas.

3) Ablation of MCEFM: Our MCEFM normalizes the high-frequency details and low-frequency context information captured by DAFM. The generated weights are then multiplied with the high-level and low-level features to integrate high-frequency and low-frequency information. During ablation, we removed the DAFM and the weight generation process, directly upsampling the high-level features and adding them to the low-level features. The experimental results validate the effectiveness of MCEFM.

4) Ablation of SSA: Introducing the attention mechanism into the model helps guide the model to focus on areas of interest, thereby enabling more precise detection of change areas. We introduced CA attention in SSA to enhance the model's ability to represent spatial features, allowing it to better focus on the change areas. During the ablation experiment, we directly removed the SSA module and compared it with MCECF. As shown in Table III, the evaluation metrics on all three datasets decreased to varying degrees. Specifically, on the WHU-CD dataset, IoU, F1, and OA decreased by 2.78%, 1.62%, and 0.26%, respectively. This indicates that introducing attention into the model can improve its ability to perceive change areas.

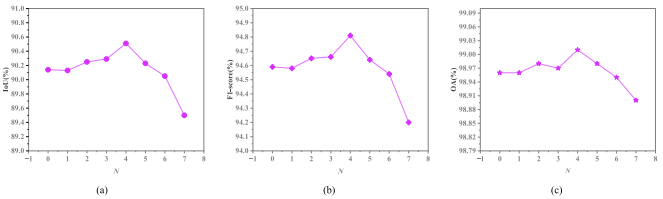


Fig. 11. Performance of MCECF with different numbers of SSA modules N on the LEVIR-CD dataset in terms of (a) IoU, (b) F1, and (c) OA.

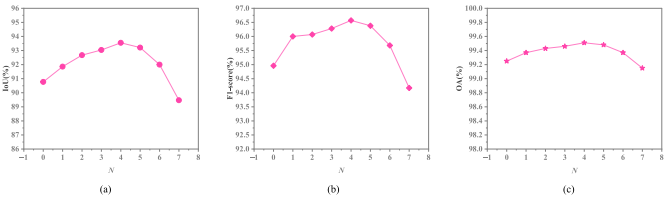


Fig. 12. Performance of MCECF with different numbers of SSA modules N on the WHU-CD dataset in terms of (a) IoU, (b) F1, and (c) OA.

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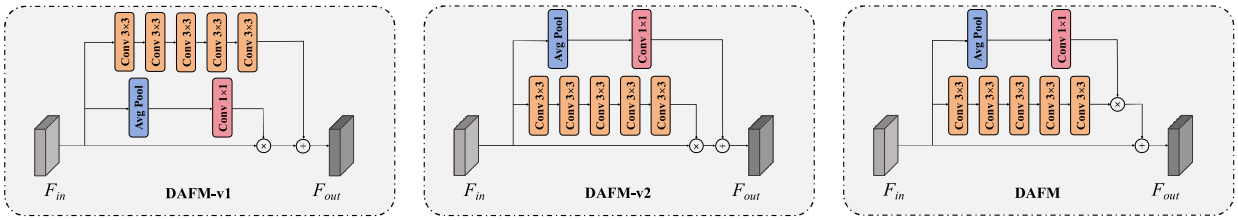


Fig. 13. Structure diagram of DAFM and its two variants. DAFM-v1 multiplies the attention branch with the input and then adds it to the convolutional branch. DAFM-v2 first multiplies the convolutional branch with the input, then adds it to the attention branch.

To investigate the effect of the number of SSA modules on the model's detection ability, we set up eight different configurations corresponding to SSA quantities ranging from 0 to 7. We evaluated the model performance for each configuration using IoU, F1-score, and OA metrics on the LEVIR-CD and WHU-CD datasets. The results are shown in Figs. 11 and 12. As the number of SSA modules increases, the model performance shows a gradual improvement, but this improvement starts to decline after the number of SSA modules reaches four. The experiments demonstrate that increasing the number of SSA modules can improve model performance to a certain extent, but beyond a critical threshold, too many modules may introduce redundant information or lead to overfitting.

5) Ablation of DAFM: DAFM dynamically adjusts the attention given to features, effectively suppressing interference and improving the model's robustness. We directly removed DAFM from the final stage but retained the DAFM in MCEFM for the ablation experiment. After removing DAFM, the evaluation metrics of MCECF decreased across all three datasets. On the LEVIR-CD dataset, IoU, F1, and OA decreased by 0.46%, 0.27%, and 0.05%, respectively. on the WHU-CD dataset, IoU, F1, and OA decreased by 1.58%, 0.92%, and 0.15%, respectively. On the CDD-CD dataset, the three metrics decreased by 0.57%, 0.31%, and 0.13%, respectively. This demonstrates that DAFM can effectively adjust the model's attention to features and suppress interference.

a) Variant comparison: Along with the ablation studies described above, we also conducted comparative experiments focused on DAFM, where we examined two alternative connection methods for DAFM's convolutional and attention branches with the input. As shown in Fig. 13, we named the two alternative connection methods of DAFM as DAFM-v1 and DAFM-v2. DAFM-v1 multiplies the attention branch with the input and then adds it to the convolutional branch.

This method significantly enhances important regions while suppressing background interference. DAFM-v2 first multiplies the convolutional branch with the input and then adds it to the attention branch. This method enhances local features while supplementing global information. We incorporated DAFM and these two variants into our MCECF, and their detection performance was quantitatively evaluated using only IoU and F1 metrics. The final results are shown in Table IV. On the LEVIR-CD dataset, DAFM-v1 decreases by 0.21% and 0.13% in IoU and F1, respectively, while DAFM-v2 decreases by 0.36% and 0.21%. On the WHU-CD dataset, DAFM-v1 shows only a slight decrease of 0.13% in F1 compared to our DAFM. However, DAFM-v2 shows a significant decline in IoU and F1, with reductions of 7.23% and 4.41%, respectively. These results indicate that DAFM is better suited for the proposed MCECF, leading to improved detection performance.

TABLE IV

QUANTITATIVE COMPARISON OF THE CD RESULTS FOR DAFM AND ITS TWO VARIANTS ON THE LEVIR-CD AND WHU-CD DATASETS

Methods	LEVIR-CD(%)		WHU-CD(%)	
	IoU	F1	IoU	F1
MCECF w DAFM-v1	90.30	94.68	93.33	96.45
MCECF w DAFM-v2	90.15	94.60	86.32	92.17
MCECF w DAFM	**90.51**	**94.81**	**93.55**	**96.58**

TABLE V

QUANTITATIVE VALIDATION OF DAFM'S PERFORMANCE IN DETECTING SMALL OBJECTS ON THE LEVIR-CD DATASET WITH A RESOLUTION OF 1024 × 1024 PIXELS

Methods	LEVIR-CD(%)_1024 × 1024				
	IoU	F1	Rec	Pre	OA
w/o DAFM	77.13	85.57	83.14	88.47	97.60
w DAFM	**80.60**	**88.23**	**86.27**	**90.44**	**98.01**

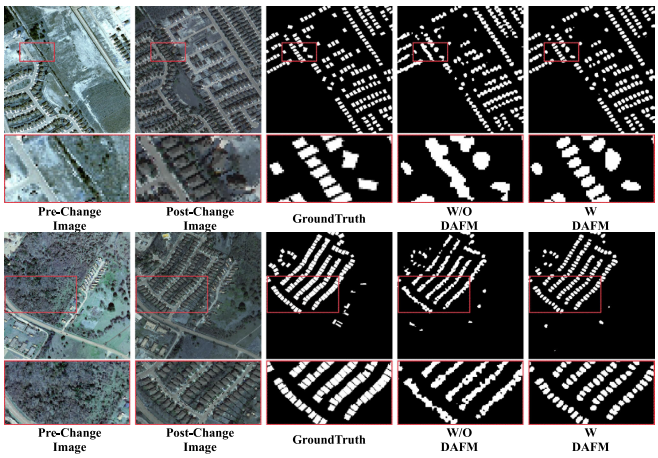


Fig. 14. Using the original LEVIR-CD dataset (1024 × 1024 resolution), two sets of test images are zoomed in to validate the effectiveness of DAFM in small-object detection.

b) Visualization of detection results for small objects: To assess the effectiveness of the proposed DAFM for small-object detection, we compressed the original 1024 × 1024 resolution images from the LEVIR-CD dataset to 256 × 256 pixels and retrained the model. We then compared the

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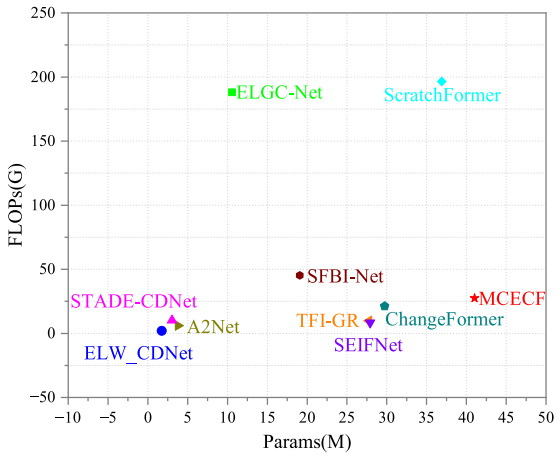


Fig. 15. Comparison of model complexity.

performance of the model with and without DAFM to validate its impact. Quantitative results, shown in Table V, indicate improvements across all five evaluation metrics after integrating DAFM. For a clearer visual comparison, Fig. 14 presents local zoomed-in change maps from both methods. The results show that adding DAFM to our MCECF enhances the accuracy of small object localization and reduces object merging in the detection outcomes.

E. Limitations

Although our method achieves good detection results compared to other SOTA methods on five datasets, there are still some limitations. First, our method is not fully effective in capturing the edge details of change areas. As a result, although it can accurately locate the changed targets in the change map, it loses edge details, as shown in Figs. 9 and 10. This may be due to the lack of edge information integration when capturing feature differences. Second, the detection of targets in complex scenes is not optimal, and there are still some FPs and FNs. Finally, compared to other methods, as shown in Fig. 15, although the FLOPs are not excessively high, the number of parameters is significantly higher than in other SOTA methods, resulting in longer training times. In future work, we will first consider combining CEDM and MCEFM, integrating differential feature extraction and multiscale feature fusion to reduce the number of parameters while maintaining the current detection performance, to reduce computational costs.

V. Conclusion

This article introduces a novel method, MCECF, for RSCD. First, the encoder block, constructed with GLCAM, captures multiscale features and global–local contextual information. Unlike the difference modules in most methods, we developed CEDM to enhance the differential information of dual-temporal features while suppressing irrelevant information. Subsequently, a progressive decoder, composed of MCEFM and SSA, is employed to integrate the abstract information from high-level features with the physical attributes of low-level features, facilitating multiscale complementary enhancement. Finally, considering the imbalance between change and nonchange regions, we propose the DAFM to enhance sensitivity to change areas. This module is also integrated into MCEFM to generate higher-quality change maps. Comparative experiments were conducted across five datasets using nine recent SOTA methods, and the results validate the effectiveness of the proposed MCECF approach.

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